Predicting Solar-Readiness of NYC Municipal Buildings

DATA 606 Final Project | Qingquan Li

NYC Open Data | City of New York Municipal Solar-Readiness Assessment (Local Law 24 of 2016) data

Abstract

New York City mandates biennial solar-readiness assessments of municipal buildings under Local Law 24 of 2016. This study examines whether observable building characteristics — roof condition, roof age, building size (gross square footage), and borough — are statistically associated with a building's solar-readiness determination.

Using 3,297 assessed NYC municipal buildings from the 2020 and 2022 reporting cycles, we fit a logistic regression model with solar-readiness (Yes/No) as the binary response variable.

Results show that (1) roof condition and building size are the strongest predictors. Buildings with "Good" roof condition have 4.2x the odds of being solar-ready compared to those with "Poor" roofs ($p < 0.001$). (2) Each unit increase in log(sqft) doubles the odds of solar-readiness ($OR = 2.02$, $p < 0.001$). (3) The Bronx has 3x the odds of Manhattan ($p < 0.001$).

Overview

Data Context

Source: NYC Open Data, City of New York Municipal Solar-Readiness Assessment (Local Law 24 of 2016) data

Collection: Biennial census of city-owned buildings > 10,000 sqft

Study type: Observational (no experimental manipulation)

Cases: 3,297 assessed buildings (2020 & 2022 cycles)

Variables

Dependent (Response):

Solar-Readiness (Yes/No) — categorical, binary

Independent (Explanatory):

1. Roof Condition (ordinal categorical)
2. Roof Age (ordinal categorical)
3. Gross Square Footage (continuous)
4. Borough (nominal categorical)

Research Question: Is there a statistically significant association between a NYC municipal building's roof condition, roof age, building size, and borough — and whether the building is assessed as solar-ready?

Summary Statistics

3,297
Assessed Buildings

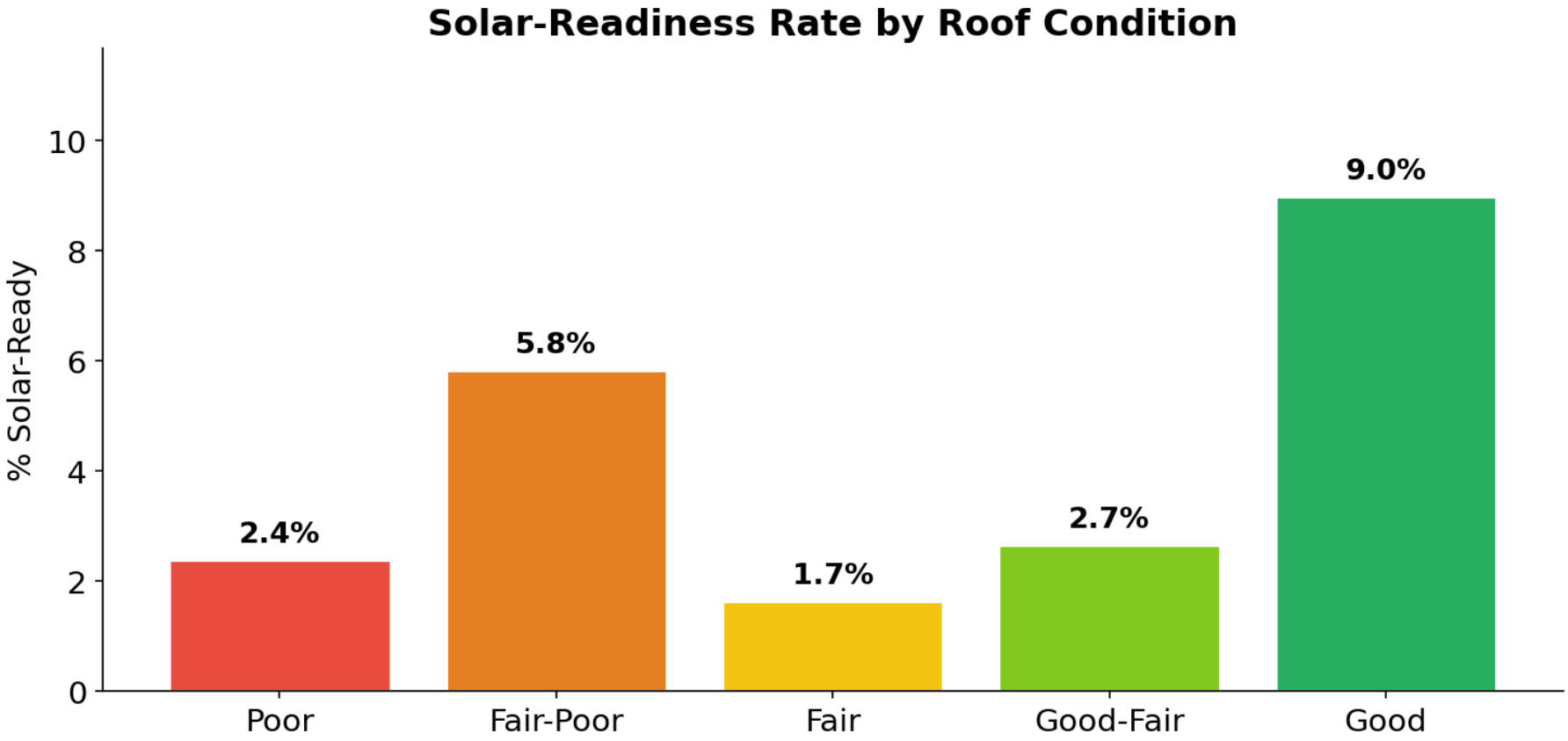
3.8%
Solar-Ready Rate (125 Solar-Ready)

104,970
Mean Sq Ft

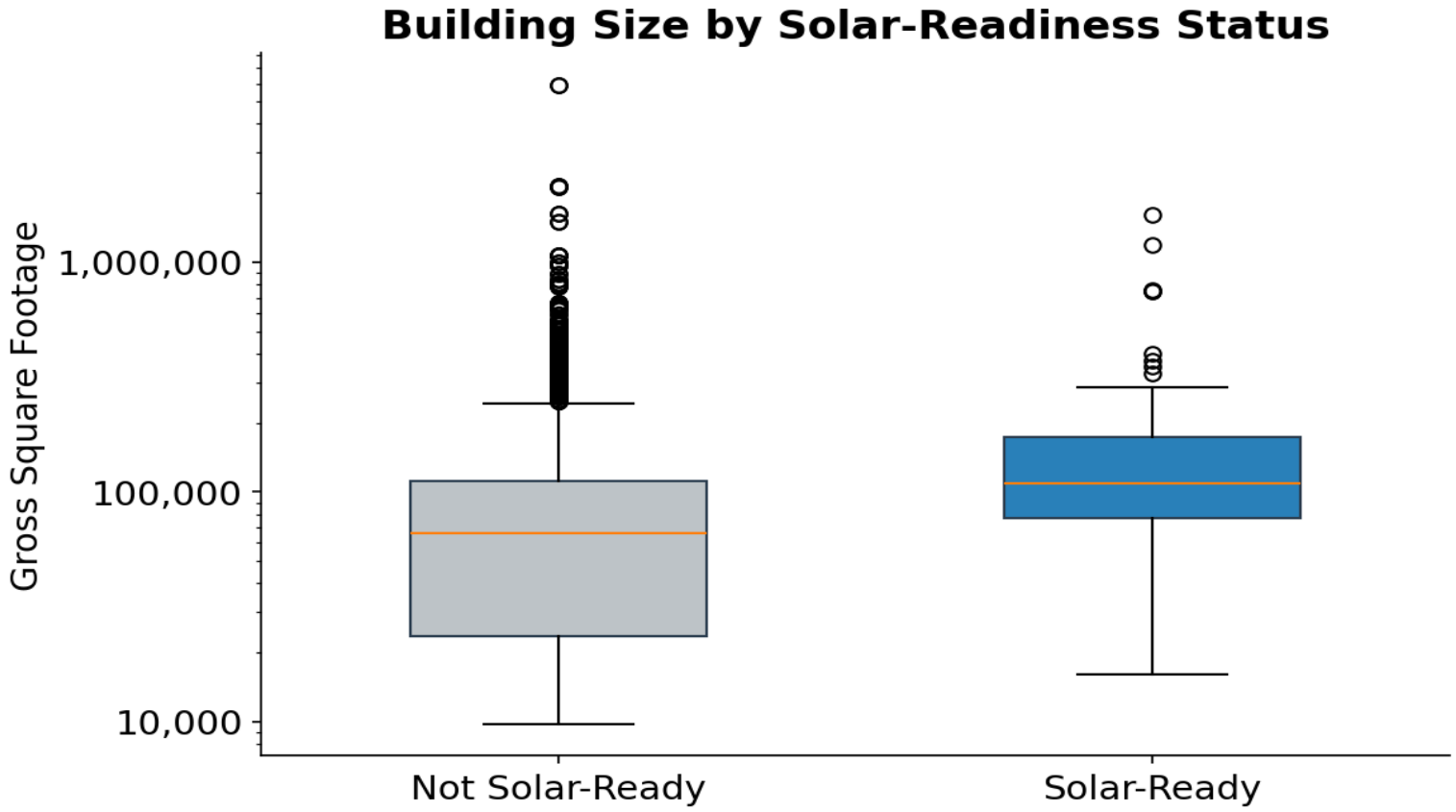
73,500
Median Sq Ft

Roof Condition	N	Solar-Ready	Rate
Poor	713	17	2.4%
Between Fair and Poor	429	25	5.8%
Fair	1,151	19	1.7%
Between Good and Fair	414	11	2.7%
Good	590	53	9.0%

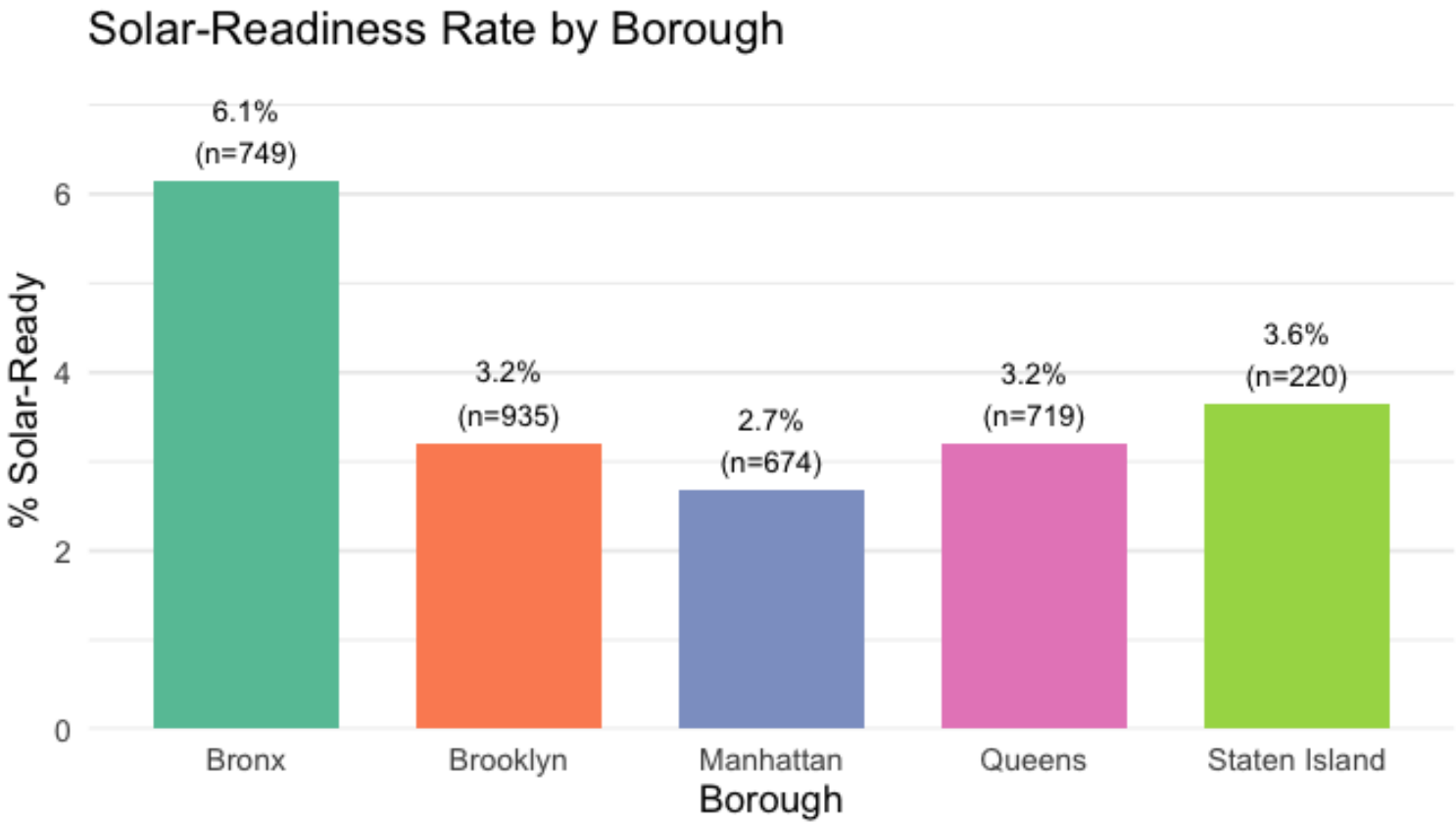
Solar-Readiness by Roof Condition



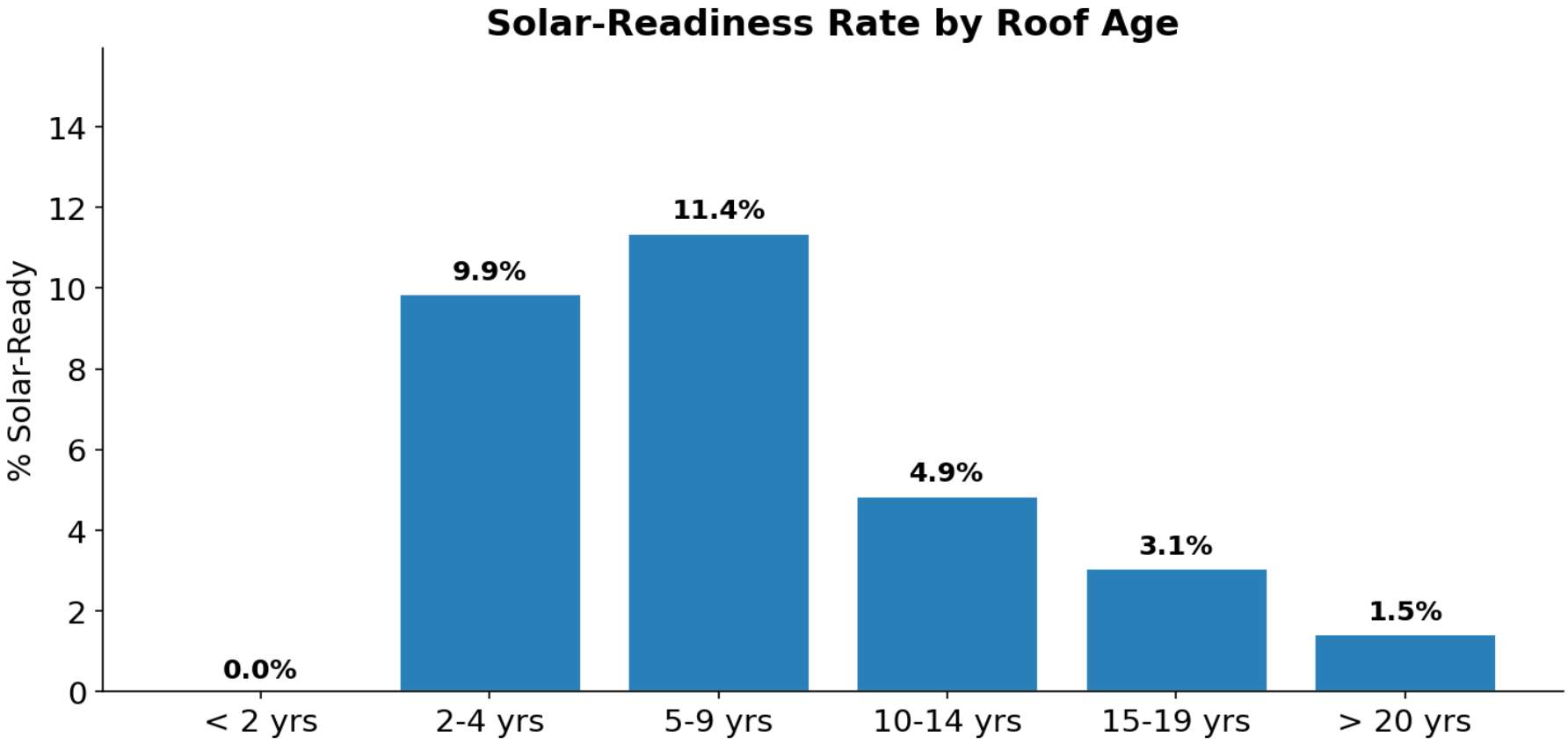
Building Size by Solar-Readiness Status



Solar-Readiness by Borough



Solar-Readiness by Roof Age



Logistic Regression Results

Model: solar_ready ~ log(sqft) + roof_condition + borough (Reference: Poor roof, Manhattan)

Predictor	Coef	Odds Ratio	z	p-value	Sig
(Intercept)	**-12.150**	**0.000**	**-10.789**	**<0.001**	***
log(sqft)	**0.702**	**2.017**	**7.887**	**<0.001**	***
Roof: Fair-Poor	**0.776**	**2.172**	**2.356**	**0.018**	*
Roof: Fair	-0.471	0.624	-1.373	0.170	
Roof: Good-Fair	0.156	1.169	0.389	0.697	
Roof: Good	**1.425**	**4.158**	**4.878**	**<0.001**	***
Boro: Bronx	**1.106**	**3.023**	**3.676**	**<0.001**	***
Boro: Brooklyn	0.415	1.514	1.305	0.192	
Boro: Queens	0.280	1.323	0.840	0.401	
Boro: Staten Isl.	0.662	1.939	1.473	0.141	

- Logistic regression — for yes/no outcomes instead of continuous numbers. It predicts the probability of "Yes" (solar-ready) based on the predictors.
- Odds ratio (e.g., 2) — for each unit increase in log(sqft), the odds of being solar-ready roughly double. Above 1 = increases the chance, below 1 = decreases it.
- p-value (e.g., 0.001) — there's less than a 0.1% chance this result is a random fluke. It's statistically significant — we're confident building size genuinely predicts solar-readiness.

Key Findings

4.2×

Buildings with Good roofs have 4.2 times the odds of being solar-ready vs. Poor roofs ($p < 0.001$).

2.0×

Each unit increase in log(sqft) doubles the odds of solar-readiness (OR = 2.02, $p < 0.001$). Larger buildings are more viable.

3.0×

The Bronx has 3× the odds of Manhattan ($p < 0.001$), possibly reflecting newer roof stock or building types.

Conclusion

Why This Matters

NYC has committed to carbon neutrality by 2050. Understanding which building characteristics predict solar-readiness helps the city prioritize roof maintenance and target capital investments to expand rooftop solar across its municipal portfolio. Improving roof conditions from Poor/Fair to Good could unlock solar potential on hundreds of additional city-owned buildings.

Limitations

1. Only 3.8% of buildings are solar-ready — severe class imbalance may affect model sensitivity.

2. Observational design — cannot establish causal relationships.

3. Only municipal buildings — results may not generalize to private buildings.

Data: NYC Open Data — City of New York Municipal Solar-Readiness Assessment (Local Law 24 of 2016)
https://data.cityofnewyork.us/City-Government/City-of-New-York-Municipal-Solar-Readiness-Assessm/cfz5-6fvh